

P4032R0

Strong ordering for `meta::info`

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Author:

[Lénárd Szolnoki](#)

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§ 1. Proposal

I propose `meta::info` to be three way comparable with an implementation-defined strong order, consistent with `std::type_order` for reflections that represent types.

§ 2. Background

[P2830R10] introduced `type_order`, which exposes an implementation-defined `consteval` strong order on types.

[P2996R13] (Reflecton for C++26) introduced `meta::info` as a structural type. This means that class template specializations can have constant template arguments of `meta::info` type. Therefore arbitrary reflection values are subject to ordering through indirection to class template specializations:

```
template <std::meta::info> struct S {};  
  
// what comes first? int or the global namespace?  
constexpr bool b = std::type_order<S<^int>, S<^::>>::value;
```

§ 3. Motivation

Being able to compare `meta::info` directly makes metaprogramming that needs to sort types, functions, etc... into some canonical order with standard algorithms more convenient.

Consider `type_set`, one of the motivating examples of [P2830R10]:

With <code>std::type_order</code> (status quo)	With <code>std::type_order</code> (proposed)
<pre>struct type_less { constexpr bool operator()(std::meta::info lhs, std::meta::info rhs) const noexcept { auto ordering_info = static_data_members_of(substitute(^std::type_order, { substitute(^S, {std::meta::reflect_constant(lhs)}), substitute(^S, {std::meta::reflect_constant(rhs)}), }), std::meta::access_context::unprivileged())[0]; return extract<const std::strong_ordering&>(ordering_info) < 0; } }; template <typename ...> struct type_set_impl; template <typename ...Ts> using type_set = [:substitute(^std::type_set_impl, std::set<meta::info, type_less>{^Ts...})];</pre>	<pre>template <typename ...> struct type_set_impl; template <typename ...Ts> using type_set = [:substitute(^std::type_set_impl, std::set<meta::info, type_less>{^Ts...})];</pre>

Moreover an ordering over reflections generalizes to cases where we want to order something other than types. For example getting a canonical order of annotations over a particular entity, instead of the lexical order.

§ 4. Implementation

[P2830R10] argues that any `operator<=>(meta::info, meta::info)` should be consistent with `type_order`. I agree.

Implementing such a comparison between arbitrary `meta::info` values is possible, based on an existing `type_order` implementation:

```
template <std::meta::info> struct _Helper {};

constexpr std::strong_ordering compare(std::meta::info a, std::meta::info b) {
    if (is_type(a) && is_type(b)) {
        // ensure that on types meta::info ordering is consistent with type_order
        auto ordering_info = static_data_members_of(substitute(
            ^std::type_order, {a, b}
        ), std::meta::access_context::unprivileged())[0];
        return extract<const std::strong_ordering&>(ordering_info);
    } else if (!is_type(a) && !is_type(b)) {
        // indirect through helper class template for non-type reflections
        auto ordering_info = static_data_members_of(substitute(
            ^std::type_order, {
                substitute(^_Helper, {std::meta::reflect_constant(a)}),
                substitute(^_Helper, {std::meta::reflect_constant(b)}),
            }
        ), std::meta::access_context::unprivileged())[0];
    }
}
```

```

        return extract<const std::strong_ordering&>(ordering_info);
    } else {
        // non-types compare less than types
        return is_type(a) <=> is_type(b);
    }
}

```

I have no implementation of the same comparison for built-in `operator<=>` in a compiler.

§ 5. Wording

Wording is relative to [N5032].

§ 5.1. [basic.fundamental]

...

- 19 There is an *implementation-defined strict total order of reflections*, such the reflection values that compare unequal (`(expr.eq)`) are also inequivalent in this total order.
- [Note 1: `int`, `const int` and `int` compare unequal to each other. - end note]
 - [Note 2: This ordering need not be consistent with the one introduced by `type_info::before`, when applied to reflections that represent types. - end note]
 - [Note 3: The ordering of reflections that represent TU-local entities from different translation units is not observable, because it is impossible to form corresponding relational or three-way comparison expressions. - end note]
- 20 *Recommended practice:* The order should be lexicographical on parameter-type-lists and template argument lists when applied to reflections representing functions and template specializations.

§ 5.2. [expr.spaceship]

...

- 7 If both operands have type `std::meta::info`, the result type is `std::strong_ordering`. The result is `std::strong_ordering::less` if the first operand precedes the second operand, `std::strong_ordering::equal` if the operands compare equal, or `std::strong_ordering::greater` if the second operand precedes the first operand according to the implementation-defined strict total order of reflections ([basic.fundamental]).

§ 5.3. [expr.rel]

...

- 2 The converted operands shall have arithmetic, enumeration, `bool` pointer, or `std::meta::info` type.
- ...
- 7 If both operands (after conversions) have type `std::meta::info`, each of the operators shall yield `true` if the specified relationship is true corresponding to the implementation-defined strict total order of reflections ([basic.fundamental]), otherwise `false`.

§ 5.4. [over.built]

...

- 15 For every T, where T is an enumeration type `bool`, a pointer type, or `std::meta::info`, there exist candidate opera-

tor functions of the form

```
bool    operator==(T, T);
bool    operator!=(T, T);
bool    operator<(T, T);
bool    operator>(T, T);
bool    operator<=(T, T);
bool    operator>=(T, T);
R       operator<=>(T, T);
```

where R is the result type specified in [expr.spaceship].

- 16 For every T, where T is a pointer-to-member type `std::meta::info`, or `std::nullptr_t`, there exist candidate operator functions of the form

```
bool operator==(T, T);
bool operator!=(T, T);
```

5.5. [compare.type]

- 1 There is an implementation-defined total ordering of all types. For any (possibly incomplete) types ~~X~~ and ~~Y~~, the expression `TYPE_ORDER(X, Y)` is a constant expression ([expr.const]) of type `strong_ordering` ([emp.strongord]). Its value is `strong_ordering::less` if ~~X~~ precedes ~~Y~~ in this implementation-defined total order, `strong_ordering::greater` if ~~Y~~ precedes ~~X~~, and `strong_ordering::equal` if they are the same type.

— [Note 1: `int`, `const int` and `int&` are different types. — end note]

— [Note 2: This ordering need not be consistent with the one induced by `type_info::before`. — end note]

— [Note 3: The ordering of TU-local types from different translation units is not observable, because the necessary specialization of `type_order` is impossible to name. — end note]

```
template<class T, class U> struct type_order {
    static constexpr strong_ordering value = TYPE_ORDER(T, U)^T <=> ^U;

    using value_type = strong_ordering;

    constexpr operator value_type() const noexcept { return value; }
    constexpr value_type operator()() const noexcept { return value; }
};
```

...

- 3 *Recommended practice:* The order should be lexicographical on parameter type lists and template argument lists

5.6. [cpp.predefined]

```
#define cpp meta info order DATE-OF-ADOPTION
```

References

Informative References

[N5032]

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[P2830R10]

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